rules.json: what it is and how it works

ROTATE team notes · 2026-11-05 · status: **engine_draft_unreviewed** (not yet checked by our science translator)

In short

rules.json is the file where ROTATE keeps its crop facts. NASA data tells us what the weather was and how green the fields were. It cannot tell us how much water wheat needs, whether mungbean adds nitrogen to the soil, or how much salt a crop can take. Those facts come from published studies, and this file is where we wrote them down, each one with its source.

The engine reads it to build the three rotation cards: the farmer's current rotation and two others to compare. Every benefit and risk on a card comes from this file, with its citation.

Why we have it

- The challenge asks us to show the benefits and risks of each rotation. Satellite and weather data can't give those. We need agronomy studies for that.
- One of our own rules is that we never say NASA data shows how a rotation performed. Keeping the crop facts in a separate file, with their own citations, keeps the two kinds of evidence apart on the cards.
- In the team plan this file belongs to the science translator. We didn't want to wait for it, so Sudipta drafted it on 5 Nov with help from Claude (AI). That's why the top of the file says **engine_draft_unreviewed**, and why every card set shows a warning until the translator has checked it.

The one rule we followed

No number goes in without a quote from the source it came from. If we couldn't find a value in a source we actually opened and read, it's left as **null**. Nothing was filled in from memory and nothing was guessed.

What null does in practice:

- On the card it shows up as an information gap, for example "No sourced irrigation figure for Lentil".
- In the score it counts as nothing. In the score breakdown it shows as null, not 0. Null means "we don't know", which is different from "this is bad".
- The engine checks this rule itself. If any value has no quote that covers it, the file won't load and the error names the crop and the column.

What's inside the file

Part	What it holds
review_status, drafted, drafted_by	Whether the file has been reviewed, when it was written and by whom.
citations	The 5 sources: full reference, link and the date we read them.
seasons	The three cropping seasons of Bangladesh, their months, and which ones count as dry season.
salinity_classes	The SRDI soil salt bands in dS/m, so a class like "slight, 4–8" becomes numbers.
crops	19 rows, one per crop (plus fallow), with the numbers and the quotes behind them.

Part	What it holds
patterns	28 rotations that farmers actually use, with their share of the country's cropland.

The three seasons

Bangladesh has three cropping seasons. The month ranges are from Emran et al. 2022. Every pattern is written in this order, which is how Nasim et al. 2017 write them: Rabi first, then Kharif-1, then Kharif-2.

id	Season	Months	Dry season?
rabi	Rabi (winter, Boro season)	mid-November to mid-March	yes
kharif1	Kharif-1 (pre-monsoon, summer)	mid-March to mid-July	yes
kharif2	Kharif-2 (monsoon, Aman season)	mid-July to mid-November	no

Two things to remember:

- **The season belongs to the slot, not the crop.** The same crop can be grown in different seasons. Mungbean is the winter crop in Mungbean–Fallow–T.Aman and the summer crop in Wheat–Mungbean–T.Aman. Boro planted after potato or mustard sits in the Kharif-1 slot.
- **Rabi and Kharif-1 count as dry season.** The salt check and the water score only look at those two slots. Nasim et al. say "the problem of salinity increases in the dry season", and the monsoon rain washes salt out. Counting Kharif-1 as dry is our own reading, so the translator should confirm it.

Soil salt classes

Class id in the file	Soil salt (dS/m)
non_saline_lt2	0–2
very_slight_2_4	2–4
slight_4_8	4–8
moderate_8_12	8–12
strong_12_16	12–16
very_strong_gt16	above 16
unknown	farmer doesn't know; no salt filter is applied

These are the SRDI 2010 bands as printed in Debsharma et al. 2024. We didn't read the SRDI report itself. The farmer (or the officer helping them) picks the class. No NASA product measures salt in a single field.

How a crop row works

Here is the wheat row, shortened. Every crop row has the same fields.


```
{ "crop_id": "wheat", "crop": "Wheat", "kind": "cereal",
  "salinity_whole_season_max_dS_m": 6.0, "salinity_seedling_dS_m": 4.0,
  "irrigation_mm": [115, 189], "n_fix_kg_ha": null,
  "evidence_strength": "weak",
  "benefits": [ { "attribute": "water", "text": "Received 115–189 mm ...", "citation_id": "hoque2023" }
],
  "risks": [ { "text": "Less salt-tolerant at seedling stage ...", "review_need": "high",
"citation_id": "fao29" } ],
  "citations": [ { "citation_id": "fao29", "columns": ["salinity_whole_season_max_dS_m", ...],
"quote": "Wheat (Triticum aestivum) ... 6.0 ..." } ] }
```


Field	What it means
crop_id, crop	Short id used in patterns, and the name shown on cards.
kind	rice, cereal, legume, oilseed, fibre, tuber, vegetable or fallow. Used to pick one legume option.
hazards_fit	Hazards a source says the crop suits. Only filled where a source says so.
salinity_whole_season_max_dS_m	Soil salt up to which the source reports full yield (FAO) or still some grain (salt-tolerant Boro). **This is the only salt number used for filtering.**
salinity_seedling_dS_m	Salt tolerance at the seedling stage. Shown for information only, never used to filter. Seedling figures are much higher and would mislead.
irrigation_mm	Irrigation water applied over the season, from a field trial.
n_fix_kg_ha	Nitrogen the crop adds to the soil, kg per hectare.
n_fert_extra_kg_ha	Nitrogen fertilizer farmers put on that crop.
organic_matter, pest_break, forage, income_note, infiltration, duration_days	All null for now. Kept so the translator can fill them later.
varieties	Variety class only, e.g. "BRRI salt-tolerant Boro variety". Named varieties stay off the cards until checked against BRRI or BARI.
evidence_strength	**strong**: a standard reference or a direct trial. **weak**: one study, or a study from another place. **principle-only**: general knowledge, no number.
benefits	Ready-to-show sentences, each with a citation_id. These go straight onto the card.
risks	Sentences with a review level (high, medium, low). A risk with citation_id null is a stated gap, not a claim.
citations	The proof. Which source, which columns it covers, and the exact quote.

How a pattern row works


```
{ "pattern_id": "potato-boro-t_aman", "name": "Potato–Boro–T.Aman",
  "slots": ["potato", "boro", "t_aman"], "seasons": ["rabi", "kharif1", "kharif2"],
  "nca_pct": 2.106, "citation_id": "nasim2017", "quote": "008 Potato–Boro–T. Aman 180380 2.106 33 115"
}
```


slots and seasons line up one to one: potato in Rabi, Boro in Kharif-1, T.Aman in Kharif-2. nca_pct is the share of the country's net cropped area under that pattern in 2014–15, straight from Nasim et al. The quote is the table row it came from.

The sources

id	Source	What we took from it
nasim2017	Nasim M, Shahidullah SM, Saha A, Muttaleb MA, Aditya TL, Ali MA, Kabir MS (2017). Distribution of Crops and Cropping Patterns in Bangladesh. Bangladesh Rice Journal 21(2): 1-55.	The list of real cropping patterns and their land share, the season order, which season each crop is grown in, and the "salinity increases in the dry season" line.

id	Source	What we took from it
fao29	FAO. Water quality for agriculture. FAO Irrigation and Drainage Paper 29, Chapter 3 (Salinity problems), Tables 4 and 5 (adapted from Maas and Hoffman 1977 and Maas 1984).	Whole-season salt tolerance (full-yield values) for rice, wheat, maize, potato, soybean, onion, chilli and others. Sesame only as "sensitive".
debsharma2024	Debsharma SK, Rahman MA, Khatun M, Disha RF, Jahan N, Quddus MR, Khatun H, Dipti SS, Ibrahim M, Iftekharuddaula KM, Kabir MS (2024). Developing climate-resilient rice varieties (BRRI dhan97 and BRRI dhan99) suitable for salt-stress environments in Bangladesh. PLoS ONE 19(1): e0294573. doi:10.1371/journal.pone.0294573	Salt-tolerant Boro lines yield at 8–10 dS/m, seedling 14 dS/m. Also the SRDI salt classes.
emran2022	Emran S-A, Krupnik TJ, Aravindakshan S, Kumar V, Pittelkow CM (2022). Impact of cropping system diversification on productivity and resource use efficiencies of smallholder farmers in south-central Bangladesh: a multi-criteria analysis. Agronomy for Sustainable Development 42: 78. doi:10.1007/s13593-022-00795-3	Season months. Mungbean adds 25–40 kg N/ha. Farmers use 11–17 kg N/ha on legumes against 60–120 extra for Boro. Coastal planting problems.
hoque2023	Hoque MA, Gathala MK, Timsina J, Ziauddin MATM, Hossain M, Krupnik TJ (2023). Reduced tillage and crop diversification can improve productivity and profitability of rice-based rotations of the Eastern Gangetic Plains. Field Crops Research 291: 108791. doi:10.1016/j.fcr.2022.108791	Irrigation applied in one trial at Jamalpur: Boro 1,186–1,795 mm, maize 254–351 mm, wheat 115–189 mm.

We also tried Mainuddin et al. 2015 (dry-season irrigation needs), Herridge et al. 2008 (nitrogen fixation) and Saraf et al. (lentil nitrogen). We couldn't open any of them (paywall or blocked), so nothing from them is in the file.

What's in the file right now

Crops

Crop	Kind	Full-yield salt (dS/m)	Irrigation (mm)	N added (kg/ha)	Evidence
Boro rice	rice	3	1186–1795	not sourced	strong
Boro rice, BRRI salt-tolerant variety	rice	8	1186–1795	not sourced	strong
Wheat	cereal	6	115–189	not sourced	weak
Maize	cereal	1.7	254–351	not sourced	weak
Mungbean	legume	not sourced	not sourced	25–40	weak
Grasspea (lathyrus)	legume	not sourced	not sourced	not sourced	weak
Mustard (short duration)	oilseed	not sourced	not sourced	not sourced	principle-only
Groundnut	legume	3.2	not sourced	not sourced	weak
Aus rice	rice	3	not sourced	not sourced	weak
Jute	fibre	not sourced	not sourced	not sourced	principle-only
Potato	tuber	1.7	not sourced	not sourced	weak
Sweet potato	tuber	1.5	not sourced	not sourced	weak
Lentil	legume	not sourced	not sourced	not sourced	principle-only
Soybean	legume	5	not sourced	not sourced	weak
Sesame	oilseed	not sourced	not sourced	not sourced	weak
Onion	vegetable	1.2	not sourced	not sourced	weak
Chilli	vegetable	1.5	not sourced	not sourced	weak
T. Aman rice	rice	not sourced	not sourced	not sourced	principle-only

Patterns (largest first)

Pattern	Crop in each season	% of cropland
Boro–Fallow–T.Aman	Rabi: Boro rice · Kharif-1: Fallow · Kharif-2: T. Aman rice	26.919
Boro–Fallow–Fallow	Rabi: Boro rice · Kharif-1: Fallow · Kharif-2: Fallow	13.302
Fallow–Fallow–T.Aman	Rabi: Fallow · Kharif-1: Fallow · Kharif-2: T. Aman rice	5.947
Boro–Aus–T.Aman	Rabi: Boro rice · Kharif-1: Aus rice · Kharif-2: T. Aman rice	2.44
Fallow–Aus–T.Aman	Rabi: Fallow · Kharif-1: Aus rice · Kharif-2: T. Aman rice	2.256
Mustard–Boro–T.Aman	Rabi: Mustard (short duration) · Kharif-1: Boro rice · Kharif-2: T. Aman rice	2.155
Potato–Boro–T.Aman	Rabi: Potato · Kharif-1: Boro rice · Kharif-2: T. Aman rice	2.106
Wheat–Jute–T.Aman	Rabi: Wheat · Kharif-1: Jute · Kharif-2: T. Aman rice	1.718
Grasspea–Fallow–T.Aman	Rabi: Grasspea (lathyrus) · Kharif-1: Fallow · Kharif-2: T. Aman rice	1.262
Maize–Fallow–T.Aman	Rabi: Maize · Kharif-1: Fallow · Kharif-2: T. Aman rice	1.184
Wheat–Fallow–T.Aman	Rabi: Wheat · Kharif-1: Fallow · Kharif-2: T. Aman rice	1.061
Mungbean–Fallow–T.Aman	Rabi: Mungbean · Kharif-1: Fallow · Kharif-2: T. Aman rice	1.047
Grasspea–Aus–T.Aman	Rabi: Grasspea (lathyrus) · Kharif-1: Aus rice · Kharif-2: T. Aman rice	0.953
Onion–Jute–T.Aman	Rabi: Onion · Kharif-1: Jute · Kharif-2: T. Aman rice	0.633
Mungbean–Aus–T.Aman	Rabi: Mungbean · Kharif-1: Aus rice · Kharif-2: T. Aman rice	0.627
Chilli–Fallow–T.Aman	Rabi: Chilli · Kharif-1: Fallow · Kharif-2: T. Aman rice	0.619

Pattern	Crop in each season	% of cropland
Lentil–Jute–T.Aman	Rabi: Lentil · Kharif-1: Jute · Kharif-2: T. Aman rice	0.606
Potato–Maize–T.Aman	Rabi: Potato · Kharif-1: Maize · Kharif-2: T. Aman rice	0.56
Boro–Jute–T.Aman	Rabi: Boro rice · Kharif-1: Jute · Kharif-2: T. Aman rice	0.472
Groundnut–Fallow–T.Aman	Rabi: Groundnut · Kharif-1: Fallow · Kharif-2: T. Aman rice	0.341
Potato–Jute–T.Aman	Rabi: Potato · Kharif-1: Jute · Kharif-2: T. Aman rice	0.33
Soybean–Fallow–T.Aman	Rabi: Soybean · Kharif-1: Fallow · Kharif-2: T. Aman rice	0.276
Wheat–Mungbean–T.Aman	Rabi: Wheat · Kharif-1: Mungbean · Kharif-2: T. Aman rice	0.25
Sweet potato–Fallow–T.Aman	Rabi: Sweet potato · Kharif-1: Fallow · Kharif-2: T. Aman rice	0.2
Lentil–Fallow–T.Aman	Rabi: Lentil · Kharif-1: Fallow · Kharif-2: T. Aman rice	0.15
Lentil–Sesame–T.Aman	Rabi: Lentil · Kharif-1: Sesame · Kharif-2: T. Aman rice	0.15
Fallow–Sesame–T.Aman	Rabi: Fallow · Kharif-1: Sesame · Kharif-2: T. Aman rice	0.136
Boro (BRRI salt-tolerant variety)–Fallow–T.Aman	Rabi: Boro rice, BRRI salt-tolerant variety · Kharif-1: Fallow · Kharif-2: T. Aman rice	variant

"variant" is the salt-tolerant Boro version of Boro–Fallow–T.Aman. It isn't a separate pattern in the survey.

How the engine uses it

1. **Card 1** is the farmer's current rotation, picked from the patterns list. It's always shown.
2. **Candidates** are the other patterns that change something in the dry season (Rabi or Kharif-1) and have at least one real crop there.
3. **Salt check.** If the farmer's salt class starts above a dry-season crop's full-yield value, that pattern is dropped. The check runs for every salt class except "unknown" and non-saline. Example: for a 4–8 dS/m field, Boro (3.0), maize (1.7) and potato (1.7) drop out, wheat (6.0) stays with a warning. A crop with no salt number stays in, with a high-review risk.
4. **Score** each candidate against the farmer's priorities. Water uses irrigation compared with Boro from the same trial. Nitrogen uses n_fix_kg_ha. Salt uses the full-yield value. Things we have no data for score null.
5. **Pick two that differ in kind:** the top option with a legume, and the top option without one.
6. **Fill the cards** with the benefits, risks and gaps from the crop rows, ranked high risk first. Every card says it needs local review.

If the salt class is 8 dS/m or more, the third card becomes "Fallow, short oilseed, or local review". The most salt-tolerant sourced Boro lines only manage 8–10 dS/m, and Nasim et al. say salinity restricts dry-season crops in tidal saline land. If the farm is already fallow in the dry season, that note goes on card 1 instead, so no card repeats. The full method is written in docs/ENGINE.md.

Weak spots we know about

- **Water numbers** exist only for Boro, wheat and maize, all from one trial at Jamalpur. That's neither of our demo sites.
- **Nitrogen** is sourced only for mungbean, and that figure is one Emran et al. quote from older studies.
- **FAO salt values are general guide values**, not Bangladeshi varieties. FAO itself says they are a guide to how crops compare, not exact limits.

- **Two kinds of salt number.** FAO gives the salt level for full yield. Debsharma gives the level where salt-tolerant Boro still produces grain. The score treats them the same, which makes salt-tolerant Boro look further ahead than it really is.
- **The salt check uses the low end of the class.** Wheat (6.0) passes a 4–8 field, with a warning that yield may fall in the saltier part. The translator should decide whether that's right.
- **Kharif-1 counted as dry season** is our reading of the sources, not a quote.
- **Dropping a summer crop scores better on water**, because fewer crops use less water. For example, Wheat–Fallow scores above Wheat–Jute.
- **No data at all** for yield stability, income, forage, equipment or pests. These always show as gaps.
- **At 8–12 dS/m**, card 2 is salt-tolerant Boro, because it is the only crop with a sourced number that high. Above 12 dS/m nothing is sourced, so card 2 falls back to an option with unknown salt tolerance.
- **Not in the file:** sunflower, vegetables as a group, felon, and two-crop patterns like Boro–B.Aman.

How to change it

- **Add a crop:** copy an existing row. Put in the numbers you have and leave the rest null. For each number, add a citations entry with the exact quote and the columns it covers. Then use the crop in at least one pattern.
- **Add a pattern:** slots and seasons must be the same length, in the order Rabi, Kharif-1, Kharif-2. Copy the survey table row into quote.
- **Add a source:** add it to citations first, with the link and the date you read it.
- **After any edit,** run `.venv/bin/python -m pytest -q engine`. If a number has no quote, or a pattern names a crop that doesn't exist, the tests fail and say where.
- **When the translator has checked everything,** change review_status to "reviewed". The warning on the cards then goes away.

Checklist for the translator

- Open each source and check every quote against it.
- Fill nulls where you can find a source: salt tolerance for mungbean, grasspea, lentil, mustard, jute and sesame; water need for everything except Boro, wheat and maize; nitrogen for grasspea, lentil, groundnut and soybean.
- Confirm or change: the low-end salt check, Kharif-1 as dry season, and the 8 dS/m cut-off.
- Check the named salt-tolerant varieties against BRRI before any name goes on a card.
- Tell us what Godagari's dry-season crop is. Its greenness peaks in February, not in March–April.

Uses NASA data. Not endorsed by NASA. This note covers the crop rule table, which is not NASA data.